

Rohan Jain

jainrohanj@gmail.com | <https://www.linkedin.com/in/rohan-jain11/> | <https://github.com/rohanjain11>
<https://rohanjain11.github.io/Portfolio/> | +1 720-819-0419 | Vista, CA

EDUCATION

University Of Colorado, Boulder

Master's in Data Science (GPA: 3.9/4.0)

Aug 2024 - May 2026 | Boulder, USA

Mumbai University, Rajiv Gandhi Institute of Technology

Bachelor's of Engineering in Computer Engineering (GPA: 3.7/4)

Jul 2020 - May 2024 | Mumbai, India

SKILLS

Languages & Data: Python, SQL (MySQL, PostgreSQL; CTEs, window functions), R, MongoDB. **Data Engineering:** ETL/ELT fundamentals, ingestion and transformation pipelines, data quality checks, schema consistency, validation gates (schema, unit, range), reproducible artifacts and documentation for reruns and handoffs. **Processing & Storage:** pandas, NumPy; Apache Spark; PostgreSQL and PostGIS for structured and spatial data workflows. **APIs & Platforms:** FastAPI, REST APIs; Git/GitHub; Docker (basics); workflow packaging and run metadata for traceable outputs. **Cloud:** AWS (Lambda, S3), Azure; practical usage for storage, compute, and service deployment. **Visualization & BI:** Tableau, Power BI, Excel (PivotTables, XLOOKUP, Power Query), Matplotlib, Plotly, Seaborn.

EXPERIENCE

Data Science & Engineering Intern, Nexus Weather & Climate

May 2025 - Present | Boulder, CO

- When forecast post-processing accuracy was limited by a legacy SVM bias-correction step, tuned and validated a Random Forest replacement and wrote rollout notes for review and promotion; reduced temperature MAE from 0.73 to 0.39 (47%) and humidity MAE from 4.68 to 2.37 (49%) on reported targets.
- To make uncertainty easier to interpret for non-ML reviewers, built a probabilistic CNN-LSTM trained with a custom CRPS loss and packaged simple verification plots and short summaries for decision reviews; achieved a best reported CRPS of 0.1787 for ensemble calibration and created a repeatable evaluation story around the result.
- When comparisons across models were inconsistent due to different inputs and evaluation paths, created a reproducible training and benchmarking workflow with schema, unit, and range checks, then ran SVR, RF, tuned RF, GBRT, XGBoost, and MLP under shared rules; produced standardized artifacts with MAE and MSE tracked against pre-ML baselines for clearer selection decisions.
- To prevent regressions from being promoted, implemented an inference gate that accepts a model only when post-ML MAE is lower than the pre-ML baseline, and made the decision automatic in the workflow; reduced failed runs by 90% by filtering out updates that did not beat baseline performance before they reached downstream review.
- When training stability and speed became a concern, ran controlled loss-function experiments where only the loss changed (CRPS vs hybrid CRPS+MSE with $\alpha = 0.80$ vs Huberized CRPS), keeping data and procedure fixed; recorded runtime and delta CRPS outcomes (3.65 min vs 4.4 min baseline, plus mean deltas) and documented practical tradeoffs for future runs.

Graduate Research Assistant, Kopf Lab, University of Colorado Boulder

Dec 2025 - Present | Boulder, CO

- When stable isotope files arrived in multiple vendor formats with inconsistent outputs, strengthened ingestion by making parsing deterministic and schema-consistent, focusing on stable field naming and unit rules; ensured the same file yields the same structured output across platforms and package versions to support repeatable downstream analysis.
- To avoid silent misparses that only appear later in analysis, implemented fail-fast validation for corruption and edge cases using type, bounds, and structural checks (missing blocks, wrong ordering, truncation); returned clear errors instead of partial or incorrect results so issues are visible at ingestion time.
- For legacy Thermo Isodat scan (.scn) files stored via MFC CArchive Serialize patterns, mapped serialized class layouts and field order from the write sequence, then implemented R readers that follow the same read order; reconstructed analysis-ready objects with traceable provenance so results are interpretable and defensible.
- To make debugging and maintenance easier for users and contributors, attached file identifiers, format signals, parsing paths, and applied assumptions to outputs, and added regression tests with golden expected outputs plus corruption fixtures; reduced the chance that reader changes break prior behavior or hide edge cases.
- To support secure multi-user scientific apps, designed a two-server hosting approach that separates authentication from app execution and configured the OIDC login flow with Keycloak for role and group access control; aligned the app layer with ShinyProxy session isolation (one container per user) to keep boundaries clearer and updates more maintainable.

PROJECTS

SaferRide (Risk-Aware Bicycle Routing) | Python, FastAPI, SQL, PostgreSQL, PostGIS,

Sep 2025 - Dec 2025

GeoPandas, OSRM, Map UI

- Built SaferRide, a routing app that requests OSRM routes, returns multiple alternatives, and scores each route using crash density and hazard proximity signals; cleaned and prepared city datasets, computed route-level risk with PostGIS spatial joins and GeoPandas, and served routes plus risk metrics via FastAPI with map overlays for safety versus time comparisons.

Denver International Airport Analytics | Python, Power BI, Tableau

Oct 2024

- Integrated ServiceNow and Azure DevOps data into analysis-ready pipelines, defined KPIs and built dashboards with drill-downs for SLA status, queue times, and bottlenecks; authored reusable ETL queries and KPI definitions for consistent refresh logic, reduced manual reporting by 40%, and earned 3rd place in a hackathon for dashboard impact.

AI PDF Chatbot | Python, FastAPI, React, LangChain, FAISS

Jan 2025

- Built a PDF question-answering tool for scanned and digital documents by combining OCR, embeddings, and FAISS retrieval behind a FastAPI and React UI; standardized chunking and prompt templates, supported secure uploads and real-time responses, added usage logging to track common queries, and used labeled spot checks to guide retrieval refinements for recurring document questions.

PUBLICATION

- Technical paper: *Yoga Posture Detection and Correction*. [Link](#)